MILITARY SPECIFICATION

ANTENNA AS-2731()/GRC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers production engineering, test and production of an asymmetrical-fed VHF whip antenna designated Antenna AS-2731()/GRC. The AS-2731 is also referenced herein as: antenna.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extend specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-S-901	Shock Test H.I.(High Impact) Shipboard Machinery, Equipment and Systems, Requirement for (Navy)
MIL-I-6869	Impregnants for Aluminum Alloy and Magnesium Alloy Castings
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-52798	Enamel, Alkyd, Camouflage

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Communication - Electronics Command, ATTN: DRSEL-[redacted]-TQ, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

HANDBOOKS

Military

MIL-HDBK-691	Adhesives

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

DL SM-B-889210	Matching Unit Base MX-9146()/GRC
DL SM-B-889249	Antenna Element AS-2732()/GRC
DL B4011671	Restraint Assembly (Safety)
DL B4011679	Adapter and Installation Hardware Kit
SC-A-46439	List of Accessories for Package Tester
DL SM-B-946377	Cable Assembly, Special Purpose, Electrical CX-13184()/GRC
GL B4001537	Mechanical gages for Restraint Assy (Safety) and Adapter

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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Electronic Industries Association Standard

RS-329 Minimum Standards for Land-Mobile Communication Antennas, Part I Base or Fixed Station Antennas, 1978

(Applications for copies should be addressed to the Electronic Industries Association, 2001 Eye Street NW, Washington, DC 20006.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified by the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data Lists DL SM-B-889210, DL SM-B-946377, DL SM-B-889249, and DL B4011671 including the requirements for parts, materials, and processes thereon. In case of conflict between the requirements of the specification and the requirements of the drawings, the drawings shall govern.

3.3 Parts, Materials, and Processes; General. In addition to the requirements of this specification, the requirement of MIL-P-11268, including selection requirement therein shall apply.

3.4 Configuration. The configuration of every Antenna AS-2731()/GRC on order shall include the following:

ITEM	QUANTITY
Antenna Element AS-2732()/GRC	1 each
Matching Unit Base MX-9146()/GRC including Mounting Hardware w/Bag.	1 each
Cable Assembly, Special Purpose, Electrical CX-13184()/GRC.	1 each
Restraint Assembly (Safety)	1 each

3.5 Electrical Characteristics. The following electrical characteristics apply to the completely assembled Antenna AS-2731()/GRC, unless otherwise stated.

3.5.1 Frequency Range. Frequency range of the antenna shall be at least from 30 to 80 MHz. It is not required that performance decline above and below this range.

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3.5.2 Sub-band Segment Bandwidth. The antenna shall comply with all the following electrical characteristics, at all frequencies within each sub-band segment for the specified ranges of frequency coverage in Table I. After band switching to each desired sub-band segment electrically or manually, no additional electrical tuning, switching, or physical adjustment shall be required.

3.5.2.1 Control Cable Voltages. The rotary solenoid switch operating voltages at the base assembly shall be as stated in Table I. The manual/automatic band switch shall operate satisfactorily and index correctly when the d-c voltage present on the band switch pins is as stated in Table I. The nominal input voltage shall be 24 Vdc. The rotary solenoid shall operate with input voltages from 20 Vdc to 30 Vdc.

TABLE I

CONTROL CABLE VOLTAGES

Sub-Band	Freq.(MHz)	AN/VRC-12 Band	Nominal Voltages at Pin N	A	B	D	E	F	H	J
1	30-33	1	24	24	0	24	0	0	0	0
2	33-37	1	24	24	0	0	24	0	0	0
3	37-42	1	24	24	0	0	0	24	0	0
4	42-47.5	1	24	24	0	0	0	0	24	0
5	47.5-53	1	24	24	0	0	0	0	0	24
6	53-56	2	24	0	24	24	0	0	0	0
7	56-60	2	24	0	24	0	24	0	0	0
8	60-65	2	24	0	24	0	0	24	0	0
9	65-70.5	2	24	0	24	0	0	0	24	0
10*	70.5-80	2	24	0	24	0	0	0	0	24

Note: Pin C is Ground - All voltages DC.

*AN/VRC-12 Band segment 10 covers 70.5 to 75.95 MHz only.

3.5.3 Free Space VSWR. Free space voltage standing wave ratio for the antenna shall not exceed those values stated in Table II for each sub-band frequency listed.

3.5.4 Antenna Impedance. The reference load impedance for all measurements, at the end of the transmission line opposite the antenna, shall be 50 ohms unbalanced to ground.

3.5.5 Matching Unit Base Alignment. The matching unit base, when aligned on the Alignment Cavity MX-9943, shall reasonate within the limits established for each sub-band as listed in Table II.

3.5.6 Antenna Gain. Antenna gain of the AS-2731 as compared to the gain of the reference standard antenna of 4.8.1 shall be no poorer than those values stated for each sub-band frequency listed in Table III. Both antennas shall be measured on a ground plane (10 ft x 10 ft copper sheet).

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TABLE II

Sub-Band Maximum VSWR and Alignment Values

Sub-band	Freq. (MHz)	Resonant Freq. Limit (MHz) for Alignment (See Note 2)	Maximum VSWR in each Sub-band (See Note 1)
1	30.0 31.5 33.0	30 $\pm$ .3	3.5
2	33.0 35.0 37.0	30 $\pm$ .3	3.5
3	37.0 42.0	30 $\pm$ .4	3.0
4	42.0 47.5	30 $\pm$ .4	3.0
5	47.5 53.0	30 $\pm$ .4	3.0
6	53.0 56.0	30 $\pm$ .4	3.0
7	56.0 60.0	30 $\pm$ .4	3.0
8	60.0 65.0	30 $\pm$ .4	3.0
9	65.0 70.5	29.8 $\pm$ .7	3.0
10	70.5 80.0	30 $\pm$ .5	3.5

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Note 1 - When measured as specified herein, maximum VSWR values listed include tolerances of test equipment and load terminations used during measurements.

Note 2 - A calibrated reference Matching Unit Base, MX-9146/GRC, should be used at the beginning of each day, or whenever there are questionable readings, to verify that the alignment cavity is operating satisfactorily.

TABLE III

AS-2731 Antenna Gain

Sub-band	Freq. (MHz)	Minimum Gain dB (See Note 1)
1	30.0 33.0	-3.5
2	33.0 37.0	-3.5
3	37.0 42.0	-3.0
4	42.0 47.5	-2.0
5	47.5 53.0	-2.0
6	53.0 56.0	-2.0
7	56.0 60.0	-2.0
8	60.0 65.0	-1.5
9	65.0 70.5	-1.5
10	70.5 80.0	-0.5

Note 1 - Antenna gain relative to Reference Standard Antenna (see 4.8.1). When measured as specified herein, gain values listed include tolerance of test equipment and load terminations used during measurements.

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3.5.7 RF Power Handling Capability. The antenna shall be capable of continuous operation with an input RF power of 70 watts for five (5) hours, without any damage to the antenna or any of its components. The reflected power shall not change by more than ± 1.5 watts throughout the test.

3.5.8 Dielectric Withstanding Voltage. Insulation material in Antenna Element AS-2732()/GRC shall show no visible effects or evidence of dielectric breakdown due to an applied voltage of 10,000 volts RMS after the antenna has been subjected to environmental stresses of the service conditions specified therein.

3.5.9 Antenna Element and Spring Assembly Continuity. Proper operation of the antenna requires the RF path from input to the spring assembly to the upper tip of the antenna element be continuous and free from undesired open or short circuits.

3.6 Mechanical Characteristics. The following mechanical characteristics shall apply:

3.6.1 Gages, Jigs and Fixtures. The contractor shall employ gages, jigs and fixtures for determining conformance with dimensional requirements.

3.6.2 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454 Requirement 7. The antenna shall be designed to provide for two-way interchangeability of parts. Selection of tolerances in the design, as shown on the drawings, shall permit any part selected from one antenna to be substituted into any other antenna of those on order without any problem of function or fit.

3.6.3 Uniformity. All units manufactured according to this specification shall be constructed from the same drawings, fabricated from the same tools and assembled to the same procedures.

3.6.4 X-ray. The antenna element shall be observed at the feedpoint and at the loading coil to determine if there is a strain relief loop at the feedpoint and to validate the location of the feedpoint and loading coil. Observation will also be made to determine that there is continuity at the antenna element feedpoint function.

3.6.5 Seal Pressure Test. The matching unit base shall be capable of withstanding an internal pressure of 15 pounds per square inch (15 pounds per square inch more than atmospheric pressure surrounding the matching unit base) for a period of 10 minutes at ambient temperature with no more than 0.03 pounds per square inch change in pressure. This is equivalent to a change in pressure of no more than 0.006 psi for a period of 2 minutes at ambient temperature.

3.7 Environmental Conditions. As a result of being subjected to the following environmental conditions, according to the methods set forth in the test schedules, the antenna shall not exhibit any effect which would be harmful to performance or reliability. The parts of the antenna shall not be deformed in any manner or exhibit evidence of corrosion. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. Harmful electrical performance is performance below the test limits stated herein.

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3.7.1 Temperature.

a. High Temperature.

(1) Operating. The antenna shall be operable without degradation in specified performance and shall sustain no physical deterioration, at ambient air temperature as high as 71° C (+160° F). This includes effects of solar radiation at a rate of 360 BTU/ft^2 /hr.

(2) Storage and Transportation. The antenna shall withstand exposure to ambient temperatures as high as 71°C (+160°F).

b. Low Temperature.

(1) Operating. The antenna shall be operable without degradation in specified performance and shall sustain no physical deterioration at ambient air temperatures as low as -45.6°C (-50°F).

(2) Storage and Transportation. The antenna shall withstand exposure to ambient temperatures as low as -57°C (-70°C).

3.7.2 Humidity. The antenna shall be operable without degradation in specified performance and shall sustain no harmful physical deterioration, as well as no evidence of water intrusion into the matching unit base during and after prolonged exposure to extreme high humidity levels encountered in tropical areas.

3.7.3 Salt Fog. The antenna shall be capable of operation for extended periods in the saline atmosphere of a seacoast environment without harmful structural deterioration, without degradation in specified performance, without insulation breakdown and without electrical flashover.

3.7.4 Dust. The antenna shall be capable of operating for extended periods in a dry, windy (17 knots) sandy environment without harmful structural deterioration, without degradation in specified performance, without insulation breakdown and without electrical flashover.

3.7.5 Immersion. The antenna, with the antenna element and matching unit base disconnected, and mounting hardware shall be capable of being immersed to a covering depth of 2 inches of fresh water for a period of 60 minutes. The absolute pressure above the liquid shall be 2.5 inches of mercury for the 60 minute period. The immersion shall not result in any form of physical deterioration, loss of fit, water or air leakage or reduction in electrical performance below specified limits.

3.7.6 Vibration and Shock. The antenna shall withstand vibration and shock induced during vehicular transport, installed within or upon a military wheeled or tracked vehicle, over all types of roads and cross country terrain, and shocks encountered in servicing. The antenna as installed upon an armored vehicle shall be resistant to high level shock caused by ballistic impact, air explosions and similar combat field conditions.

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3.7.7 Impact. The antenna element shall be capable of withstanding repeated blows from immovable obstacles when mounted on ground vehicles traveling at speeds up to 25 mph. Physical damage, if any, shall be limited to minor surface abrasion at the point, no deeper than 0.005 inches.

3.7.8 Spring and Coaxial Assembly Flexibility. The antenna spring and coaxial assembly shall be capable of withstanding the flexing experienced on ground vehicles under all types of environmental conditions outlined in this specification without electrical or mechanical damage. Flexing of the spring will be caused by ground vehicles traveling over rough terrain, through wooded areas with over-hanging branches, and under other obstacles such as bridges and overhead power or telephone lines. Under some conditions the antenna could spring back as much as 135° from the normal vertical position. The spring must survive flexing caused by the conditions above and the coaxial assembly must survive the same flexing conditions without electrical failure, broken or pinched insulation, or evidence that the coaxial cable has loosened from the coaxial connectors.

3.7.9 Temperature Shock. The antenna shall be capable of withstanding the effects of sudden changes of temperature of the surrounding atmosphere.

3.7.10 Bounce. The antenna shall be capable of withstanding the effects of bounce as tested in accordance with 4.11.4.

3.7.11 Fungus. The antenna, with matching unit case (SM-D-889211) open, shall provide no fungus nutrients in the material, coating, or contaminant form and shall not support fungal growth when tested. After being tested the antenna shall show no more than six unreleated spots, each no greater than 0.015 square inch in area, of microbial growth (see 6.11) as evidenced by growth colonization (see 6.11) (which includes branching (see 6.11) and sporulation (see 6.11)) on or within each cubic foot, or fraction thereof, of antenna volume. Isolated instances of partial tubular germination (see 6.11) shall not be included in this evaluation.

3.8 Parts, Materials, and Processes. Parts, materials and processes shall be in accordance with specification MIL-P-11268.

3.8.1 Plastic Material and Parts. Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application. Plastic material and parts shall conform to the following, except that plastic material used for arc resistance shall be limited to the types specified in Requirement 26 of MIL-STD-454. During the selection of plastic materials, consideration should be given to the tropicalization requirements of specification MIL-P-11268.

3.8.2 Adhesives. Use of adhesives shall be in accordance with Requirement 23 "Adhesives" of MIL-STD-454. MIL-HDBK-691 shall be used as a guide in the use of adhesives.

3.9 Finishes. Finishes shall conform to MIL-F-14072. All metallic, polycarbonate, epoxy and polyester glass laminate surfaces shall be painted with a

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coating material approved under MIL-E-52798 "Enamel, Alkyd, Camouflage", color Forest Green.

3.10 Marking.

3.10.1 General. Marking shall conform to MIL-M-13231. Identification plate marking shall be group I as described in that specification.

3.10.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.10.3 Serial Numbers. The Matching Unit Base MX-9146()/GRC, shall be serial numbered. The serial numbers should be capable of isolation of a given unit to a given production lot/run.

3.10.4 National Stock Numbers. The Matching Unit Base MX-9146()/GRC, Antenna Element AS-2732()/GRC, Cable Assembly CX-13184, and Restraint Assembly shall be marked with their applicable National Stock Numbers.

3.11 Systems Safety Engineering.

3.11.1 General. Safety requirements shall be in accordance with Requirement 1, MIL-STD-454 and the paragraph entitled "Hazards and Safety" of MIL-STD-1472.

3.11.2 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.12 Human Engineering. The antenna shall conform to Requirement 62 of MIL-STD-454.

3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following:

a. In MIL-P-11268

Inspection

Wire hookup
Cleaning
Finish, protective
Rivets and Riveting
Soldering
Brazing and welding
Wiring and cabling
Controls
Screws, other threaded devices and related parts

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b. In this specification:

Paragraph	Inspection
3.8.1	Plastic materials and parts
3.10	Marking

3.14 Bounce Preconditioning. The MX-9146()/GRC shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.9 (also see 4.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article inspection (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.18).

4.3 First Article Inspection. This inspection shall consist of the inspection specified in Table IV and shall include inspection specified in subsidiary documents and the inspection specified for Group A, Group B, and Group C. Order and quantity of equipments to be subjected to First Article Testing shall be as specified in Table IV.

4.3.1 Procedure in the Event of Failure During First Article Inspection. In the event of failure to pass any First Article inspection, the unit shall be corrected as necessary and then resubmitted to First Article Inspection. After successful retest, the same corrections shall be applied to all other antennas on order.

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TABLE IV

FIRST ARTICLE TEST SCHEDULE

TEST TITLE	RQT PARA	TEST PARA	SEQUENCE UNITS 1-5	UNITS 6-10	UNITS 11-15	UNITS 16-20	UNITS 21-25
Visual Inspection	3.6.3 3.8-3.13	4.12	1	1	1	1	1
Temperature Shock	3.7.9	4.11.3	2	2	2		2
Bounce	3.7.10	4.11.4	3	3	3		3
Seal Pressure Test	3.6.5	4.13	4	4	4		4
Item Count	3.4	4.14	5	5	5		5
Mechanical	3.6	4.15	6	6	6		6
Interchangeability	3.6.2	4.16	7	7	7		7
Immersion	3.7.5	4.11.2	8	8	8		8
Free Space VSWR	3.5.3	4.10.2	9	9	9		9
Matching Unit Base Alignment	3.5.5	4.10.6	10	10	10		10
Antenna Gain	3.5.6	4.10.4	11	11	11		11
Dielectric Withstanding Voltage	3.5.8	4.10.5	12	12	12		12
X-Ray	3.6.4	4.17	13	13	13		13
RF Power Handling Capability	3.5.7	4.10.3	14	14	14		14
Control Cable Voltages	3.5.2.1	4.10.1	15	15	15		15
Inspection for Physical Damage		4.12.1					
High Temperature	3.7.1	4.11.5				2	
Low Temperature	3.7.1	4.11.6				3	
Humidity (1)	3.7.2	4.11.7				4(1)	
Salt Fog	3.7.3	4.11.9			16		
Dust	3.7.4	4.11.1	16				
Fungus (1)	3.7.11	4.11.8				4(1)	
Vibration	3.7.6	4.11.13					16
Ballistic Shock	3.7.6	4.11.10		16			
Impact	3.7.7	4.11.11		17			
Spring and Coaxial Assembly Flex (2)	3.7.8	4.11.12	(5000 flexes)		(40,000 flexes)		(500 flexes)

(1) 2 units from group for Humidity; 3 units from group for Fungus.
(2) 5 each spring assy. each test; separate base unit as test fixture for Pre and Post Tests.

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4.3.2 First Article Sample. Following acceptance of the First Article samples, two (2) units shall remain in the custody of the Government Quality Assurance Representative for his use as a reference. Following completion of production on order, the units shall be reconditioned to whatever extent necessary and then shipped according to the contract.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:

Item	Paragraph
Parts, Material and Processes used in Electronic Equipment	3.8
Marking	3.10

4.5 Quality Conformance Inspection of Equipment Before Packaging. Each unit of each lot of equipment shall be subjected to bounce preconditioning (4.9) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.1.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. Further, all facilities and services necessary for the placement, operation and maintenance of these test equipments shall be provided by the contractor. The contractor shall be responsible for providing all apparatus which may be required for qualification and acceptance tests required herein.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table V. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to Sampling Inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table V. Group A inspection shall be performed in any order satisfactory to the Government with the exception that the seal pressure test (4.13) shall be conducted after all tests except item count (4.14).

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TABLE V

GROUP A INSPECTION

TITLE	RQT PARA	INSP PARA	AQL MAJOR	AQL MINOR
Visual Inspection	3.6.3; 3.8; 3.9; 3.10; 3.11; 3.12; 3.13	4.12	1%	4%
Control Cable Voltages	3.5.2.1	4.10.1	1%	
Seal Pressure	3.6.5	4.13	1%	
Item Count	3.4	4.14	1%	
Free Space VSWR	3.5.3	4.10.2	1%	

4.5.1.1 Procedure in the Event of Failure During Group A Inspection. In the event of failure to pass any Group A inspection item, the unit shall be corrected as necessary and then resubmitted to Group A inspection. After successful retest, the same corrections shall be applied to all other undelivered antennas on order.

4.5.2 Group B Inspection. This inspection (including sampling) shall conform to Table VI, and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection level S-4. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A.

4.5.2.1 Order of Inspection Within Group B. Group B inspection shall be performed in any order which is satsifactory to the Government.

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TABLE VI

GROUP B INSPECTION

TITLE	RQT PARA	INSP. PARA.	AQL MAJOR	AQL MINOR
Mechanical	3.6	4.15	4.0%	
Bounce	3.7.10	4.11.4		
RF Power Handling	3.5.7	4.10.3		
X-Ray	3.6.4	4.17		
Immersion	3.7.5	4.11.2	4.0%	
Interchangeability	3.6.2	4.16	6.5%	
Temperature Shock	3.7.9	4.11.3	4.0%	

4.5.2.2 Procedure in the Event of Failure During Group B Inspection. In the event of failure to pass any Group B inspection item, the unit shall be corrected as necessary and then resubmitted to Group B inspection. After successful retest, the same corrections shall be applied to all other undelivered antennas on order.

4.5.3 Group C Inspection. This inspection shall consist of the tests specified in Table VII and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1. Testing of samples slected shall begin within seven (7) days after the samples have been selected.

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TABLE VII
GROUP C INSPECTION

TITLE	RQT PARA	INSP. PARA.	SUB-GROUP
High Temperature	3.7.1	4.11.5	II
Low Temperature	3.7.1	4.11.6	II
Humidity	3.7.2	4.11.7	III
Fungus	3.7.11	4.11.8	IV
Salt Fog	3.7.3	4.11.9	IV
Antenna Gain	3.5.6	4.10.4	IV
Ballistic Shock	3.7.6	4.11.10	III
Impact	3.7.7	4.11.11	I
Spring and Coaxial Assembly Flexibility	3.7.8	4.11.12	II
Vibration	3.7.6	4.11.13	II

4.5.3.1 Sampling for Inspection of Equipment. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Sub-group I. For this sub-group, two units from the first 50 production units shall be selected for this first Group C inspection. For subsequent Group C inspection, two units from each successive 250 or two per month if less than 250 are produced monthly, shall be selected.

4.5.3.1.2 Sub-group II. For this sub-group, two units from the first 50 production units shall be selected for this Group C inspection. For subsequent Group C inspections, two units from each successive 500, or two every two months, if less than 500 are produced during a two month period.

4.5.3.1.3 Sub-group III. For this sub-group, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units shall be selected from each successive 600, or two every three months, if less than 600 are produced during this three month period.

4.5.3.1.4 Sub-group IV. For this sub-group, two units from the first 50 production units only shall be selected. Subsequently, two units shall be selected from each successive 3000 units, or two every six months if less than 3000 units are produced until completion of production.

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4.6 Standard Test Conditions. Unless otherwise specified, the following standard test conditions shall prevail during testing:

Condition	Value/Tolerance
Temperature	Prevailing Ambient
Altitude	Ground Level

4.7 Configuration to be Tested. The matching unit base, antenna element, mounting hardware, adapter control cable, and restraint assembly shall be submitted together for test during First Article and Quality Conformance Tests.

4.8 Test Equipment. All electrical test equipment employed, except for the 70 watt RF power source, shall be commercially available items for which calibration service, referencing back to primary standards, is available. The 70 watt RF power source for RF power handling capability testing may be any unmodulated generator capable of adjustment to the required power level, capable of adjustment to the designated test frequency, and having a harmonic content of less than 10%.

4.8.1 Reference Standard Antenna. The reference standard antenna for gain measurements shall be a vertically polarized quarter-wave monopole. Length of the monopole shall be continuously adjustable. At each test frequency, length shall be adjusted to quarter-wave length resonance considering end effects (usually 95% to 97% of one quarter-wave length). When so adjusted, free space VSWR shall be less than 2.5 to 1. The vertical monopole shall be mounted at the center of a horizontal ground plane (10 ft x 10 ft copper sheet). A BNC connector shall be provided to attach the transmission line. The ground side of the connector shall be attached to the ground plane and the center conductor shall be attached to the vertical monopole. The transmission line shall be the same line or an identical line to that of the antenna under test. Free space VSWR of the antenna shall be measured over the frequency range of the antenna under test. Maximum VSWR shall be recorded. The ground plane for each antenna under test shall be the same ground plane used for the reference standard antenna.

4.9 Bounce Pre-conditioning. The matching unit base shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32-inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this pre-conditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.10 Electrical Tests. Unless specified elsewhere, the complete antenna system shall be subjected to the following electrical tests.

4.10.1 Control Cable Voltages. Twenty (20), twenty-four (24), and thirty (30) Vdc control voltages shall be switched to the pins stated in Table I for each sub-band. The matching unit base shall switch quickly and index correctly with no hesitation. This test shall be conducted with the adaptor control cable attached to the matching unit base.

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4.10.2 Voltage Standing Wave Ratio (VSWR) Test.

4.10.2.1 Free Space VSWR. The entire AS-2731/GRC antenna shall be mounted on a 10 ft x 10 ft ground plane under the conditions of 4.6 above. The VSWR shall be measured by any commercially standard means acceptable to the Government to determine compliance with paragraph 3.5.3 of this specification. EIA Standard RS-329 may be used as a guide.

4.10.3 RF Power Handling Capability. Seventy (70) watts of power shall be applied to the mounted antenna at the RF input connector. The power source shall be a continuous wave, radio frequency source having a nominal output impedance of 50 ohms. One test frequency from each sub-band shall be selected. Power shall be applied for a period of 30 minutes for each test frequency, and forward and reflected power shall be recorded at the beginning of the test and at 5 minute intervals. After power is removed, the insulation material shall be inspected for evidence of RF flashover or leakage. Evidence of RF flashover, leakage or variations of reflected power of values greater than $\pm$ 1.5 watts shall constitute a failure. When conducted as a post-environmental test, the test shall be completed within 10 hours of the time of completion of exposure to the environmental condition. The test may be conducted in Antenna Test Chamber MX-9930 or on a 10 ft x 10 ft ground plane when FCC approved test frequencies are available.

4.10.4 Antenna Gain. In accordance with para 2.11.3 of RS-329 except that the reference standard antenna shall conform to 4.8.1 of this specification and the antenna under test shall be full scale size and mounted on the reference standard antenna ground plane. Test frequencies shall be at least those frequency values listed in Table III for each subband and shall be sufficient so that a plot of gain vs frequency will faithfully include maximum and minimum points within that range.

4.10.5 Dielectric Withstanding Voltage. The test shall be performed on Antenna Element AS-2732()/GRC in accordance with MIL-STD-202, Method 301. The test voltage shall be 10,000 volts RMS. The test voltage shall be applied along the full length of the antenna element to within one inch of the antenna element's ferrule.

4.10.6 Matching Unit Base Alignment. The matching unit base, when inserted into Alignment Cavity MX-9943, shall resonate such that the frequency response curve falls within the limits stated in Table II. Actual resonant frequency shall be measured for each sub-band and shall be determined by noting the peak of the response curve displayed on an oscilloscope.

4.10.7 Antenna Element and Spring Assembly Continuity. The antenna element and spring assembly shall be subject to an RF test to verify that no open or short circuit malfunctions are present in the unit under test after the antenna has been subject to environmental stress of the service conditions specified therein.

4.11 Environmental Service Condition Tests.

4.11.1 Dust. The antenna system shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810. Step 3 shall be performed immediately after

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temperature has stabilized during Step 2. The following measurements shall be made before and after the test cycle. Post-tests shall be conducted without any cleaning other than due to necessary handling.

Test	Paragraph
RF Power Handling	4.10.3
Matching Unit Base Alignment	4.10.6
Dielectric Withstanding Voltage	4.10.5

4.11.2 Immersion. The antenna system shall be tested in accordance with Method 512.1, Procedure III of MIL-STD-810. The test items shall be completely immersed in water so that the uppermost part of the test items are 2 ± 1 inches below the surface of the water. The absolute pressure of the air above the water shall be reduced to 1 inch of mercury (absolute) and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is the longer time. The absolute pressure above the liquid shall then be increased to 2.5 inches of mercury and maintained for 60 minutes. Bubbles coming from within the equipment shall be considered a leakage; however, bubbles which result from trapped air on the various external surfaces of the test item shall not be considered as leaks. On removal from the water, the antenna shall be subjected to post-test procedures without being dried in any manner other than induced by necessary handling. Post tests shall be completed within 8 hours after removal. During the Inspection for Physical Damage (4.12.1) the antenna shall be checked for leakage of moisture by removing the matching unit base cover. Presence of moisture shall constitute a failure. The following tests shall be conducted as pre- and post-tests.

Test	Paragraph
Matching Unit Base Alignment	4.10.6
RF Power Handling	4.10.3
Dielectric Withstanding Voltage	4.10.5
Antenna Element & Spring Assy Continuity	4.10.7
Inspection for Physical Damage	4.12.1

4.11.3 Temperature Shock. The antenna shall be subjected to Procedure I of Method 503.1 of MIL-STD-810. During Step 1 the internal chamber temperature shall be maintained at 70°C (160°F) for 90 minutes. During Step 3 the internal chamber temperature shall be maintained at -57°C (-70°F) for 90 minutes. Step 5 shall be for 90 minutes. Delete Step 6. Conduct Steps 7, 8 and 9 using temperature and time criteria above. The following tests shall be conducted as pre- and post-tests.

Test	Paragraph
Matching Unit Base Alignment	4.10.6
Antenna Element & Spring Assy Continuity	4.10.7
Inspection for Physical Damage	4.12.1

4.11.4 Bounce. The antenna shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from

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those listed on USA CECOM Drawing SC-A-46439. As a pretest, the antenna shall be subjected to 4.10.6. The test shall be as follows:

a. Secure the equipment to the 48 by 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

b. Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space six-penny nails, with the head below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

c. Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to nor more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

d. Operate the package tester, shafts in phase, for a total of 3 hours at 283 $\pm$ 1 rpm. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/8 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces.

e. After bounce testing, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to the following post-tests:

Matching Unit Base Alignment (4.10.6) and
Inspection for Physical Damage (4.12.1).

4.11.5 High Temperature. The antenna shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. During Step 8, the matching unit base shall be subjected to the RF Power Handling Capability Test 4.10.3, for 1 hour. During this test, the matching unit base may be connected to a RF dummy load and shall be tested at two (2) frequencies (one each in sub-bands 5 and 10) for 30 minutes each. The following tests shall be conducted as pre- and post-tests.

Test	Paragraph
RF Power Handling	4.10.3
Matching Unit Base Alignment	4.10.6
Seal Pressure	4.13
Inspection for Physical Damage	4.12.1

4.11.6 Low Temperature. The antenna shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. During Step 2 the chamber temperature shall be maintained at -57°C (-70°F) for not less than 2 hours after stabilization of the test items. Delete Step 3. During Step 4, the chamber temperature shall be maintained at -45.6°C (-50°F) for not less than 4 hours after stabilization of the test items. During Step 5, the matching unit base shall be subjected to the RF Power Handling Capability Test 4.10.3 for 1 hour. During this test, the matching unit base may be connected to a RF dummy load and shall be tested at

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two frequencies (one each in sub-bands 5 and 10) for 30 minutes each. The following tests shall be conducted as pre- and post-tests:

Test	Paragraph
RF Power Handling	4.10.3
Matching Unit Base Alignment	4.10.6
Seal Presure	4.13
Inspection for Physical Damage	4.12.1

4.11.7 Humidity. The antenna shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. The antenna shall be fully assembled in the test chamber. Mounting hardware shall be arranged in the chamber so that the parts are unprotected by the bag. As a pretest, at Step 4, the antenna assembly shall be removed from the chamber and subjected to the RF Power Handling Capability 4.10.3 and Matching Unit Base Alignment 4.10.6. Following these tests, the equipment shall be replaced in the chamber. In Step 6, the matching unit base shall be subjected to RF Power Handling Capability Test, 4.10.3, for one (1) hour during the last five (5) hours of the last cycle. During this test, the matching unit base may be connected to a RF dummy load and shall be tested at two (2) frequencies (one each in sub-bands 5 and 10) for 30 minutes each. In Step 7, the matching unit base shall be opened so as to expose the sealed areas to the chamber environment. During the last five (5) hours of the 480 hour exposure in Step 8, the matching unit base shall be subjected to the RF Power Handling Capability Test, 4.10.3 for one (1) hour. During this test, the matching unit base may be connected to a RF dummy load and shall be tested at two (2) frequencies (one each in sub-bands 5 and 10) for 30 minutes each. The following post-tests shall be conducted within one (1) hour of the Step 9 Conditioning:

Dielectric Withstanding Voltage	4.10.5
Matching Unit Base Alignment	4.10.6
Inspection for Physical Damage	4.12.1

In addition, the following post-tests shall be conducted:

RF Power Handling Capability	4.10.3
Seal Pressure Test	4.13

4.11.8 Fungus. The antenna shall be tested in accordance with Method 508 Procedure I of MIL-STD-810. The antenna matching unit case (SM-D-889211) shall be opened, the AS-2732/GRC element shall not be connected to the MX-9146/GRC base, the parts and materials of the antenna internal assembly, the interior of the case, the interior surfaces of the cable mount assembly (SM-D-889199), as well as the exterior of antenna shall be sprayed with the spore suspension. The antenna shall be subjected to the test in this open condition. There shall be abundant growth colonization (see 6.11) on 50% or more of the control items after 7 and 28 days. Inability of the antenna to meet the requirements of 3.7.11 shall constitute failure of this test.

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Test	Paragraph
RF Power Handling	4.10.3
Matching Unit Base Alignment	4.10.6
Dielectric Withstanding Voltage	4.10.5
Inspection for Physical Damage	4.12.1

4.11.9 Salt Fog. The antenna shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810. The antenna shall be mounted in the chamber to receive exposure. The following tests shall be conducted as pre- and post-tests.

Test	Paragraph
RF Power Handling	4.10.3
Matching Unit Base Alignment	4.10.6
Dielectric Withstanding Voltage	4.10.5
Seal Pressure	4.13
Inspection for Physical Damage	4.12.1

4.11.10 Ballistic Shock. The antenna shall be subjected to the ballistic shock test on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The antenna shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows. The sequence of test blows is to be:

(1) 1-foot, 3-foot, and 5-foot blows on the back;
(2) 1-foot, 3-foot, and 5-foot blows on the top; and
(3) 1-foot, 3-foot and 5-foot blows on the side.

As an alternative to re-orienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. The following test shall be conducted as pre- and post-tests:

Test	Paragraph
Matching Unit Base Alignment	4.10.6
Dielectric Withstanding Voltage	4.10.5
Antenna Element & Spring Assy. Continuity	4.10.7
Inspection for Physical Damage	4.12.1
Seal Pressure Test	4.13

4.11.11 Impact. The antenna (less restraint assembly), secured on a vehicle in a normal operating position, shall be subjected to 25 repeated blows in any direction, against an oak beam having a cross-section of 4 inch x 4 inch, when the vehicle is traveling at a rate of 25 mph. The point of impact shall be at the mid-point of the Antenna Element AS-2732/GRC. The oak beam shall be stationary at the point of impact. The following tests shall be conducted as pre- and post-tests:

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Test	Paragraph
X-ray	4.17
Dielectric Withstanding Voltage	4.10.5
Antenna Element & Spring Assy. Continuity	4.10.7
Seal Pressure Test	4.13
Inspection for Physical Damage	4.12.1

4.11.12 Spring and Coaxial Assembly Flexibility. The following tests shall be performed at -40°F and the flexing rate shall be 10 cycles per minute plus or minus one cycle per minute. Flexing tests may be accomplished by rotating and cycling the spring at each of three (3) 120° positions.

NOTE: Assemblies undergoing these tests shall not be used as deliverable items.

4.11.12.1 40,000 Cycle Flex. One matching unit base shall have its spring and coaxial assembly flexed 40,000 cycles with each cycle consisting of a deflection of the top of the spring +30° from the unflexed position, to -30°, and return to normal.

4.11.12.2 5,000 Cycle Flex. A second matching unit base shall have its spring and coaxial assembly flexed 5,000 cycles with each cycle consisting of a deflection +90° from the unflexed position, to -90°, and return to normal.

4.11.12.3 500 Cycle Flex. A third matching unit base shall have its spring and coaxial assembly flexed 500 cycles with each cycle consisting of a deflection +135° from the unflexed position, to -135°, and return to normal. Breakage of the spring does not constitute a failure of this test. The following tests shall be conducted as pre- and post-tests:

Test	Paragraph
RF Power Handling (75.75 MHz for 30 minutes)	4.10.3
Inspection for Physical Damage	4.12.1

4.11.13 Vibration. The antenna and all mounting hardware, shall be subjected to the test of Method 514.2, Procedure VIII, Curve W of Figure 514.2-6 of MIL-STD-810. The following tests shall be conducted as pre-tests and post-tests:

Test	Paragraph
RF Power Handling Capability	4.10.3
Matching Unit Base Alignment	4.10.6
Seal Pressure Test	4.13
Inspection for Physical Damage	4.12.1

4.12 Visual Inspection. The equipment shall be visually inspected for compliance with the requirements of this specification, especially para. 3.13 Workmanship. Classification of defects shall be per MIL-STD-252.

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4.12.1 Inspection for Physical Damage. As a result of being subjected to environmental conditions, the antenna shall not exhibit any effect which would be harmful to performance or reliability. The test items shall be visually inspected for evidence of physical damage such as corrosion, deformation, and cracked or broken parts.

4.13 Seal Pressure Test. The matching unit base shall be opened and closed again in such manner as to break and re-make the seal. The matching unit base shall then be subjected to an internal pressure of 15 psi (15 psi more than the atmospheric pressure surrounding the matching unit base) $\pm$ 0.25 psi, utilizing nitrogen gas. The pressure then shall be valved-off and the internal pressure measured during the ensuing period of 2 minutes. During this 2 minute period, there shall be no more than 0.006 psi change in pressure. The gage used for measurement of the pressure shall be of such accuracy that any change in pressure of 0.001 psi can be readily determined. One of six screws of the side cover shall be removed and used to install a pressure fitting. The screw shall be replaced at completion of the test, without affecting the seal.

4.14 Item Count. Each item required to complete the antenna assembly shall be inventoried as it is placed in its assigned position in the bag and shipping container.

4.15 Mechanical. Thread engagement and dimensional control tests shall be conducted with appropriate gages, jigs, or fixtures. The restraint assembly (safety) and adapter shall be gaged using gages shown on GL-B4001537.

4.16 Interchangeability. Within each system all parts to be fitted together shall be assembled or fitted together. Observation for correctness of fit shall be conducted. These parts shall then be interchanged among other samples (as designated by the Government Quality Assurance Representative) and assembled or fitted together. Observation for correctness of fit in both systems shall be observed. Parts recommended to be interchanged are:

Antenna Element,
Base Spring Assembly,
Control Cable Assembly.

4.17 X-Ray. The antenna element shall be x-rayed along the feedpoint area and the loading coil area to determine compliance with para. 3.6.4.

4.18 Quality Conformance Inspection of Packaging. Packaging shall be inspected in accordance with SPI-1G00028 to determine conformance to the requirements of Section 5.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with SPI-1G00028.

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6. NOTES

6.1 Intended Use. Antenna AS-2731()/GRC is intended for use with Radio Sets AN/VRC-12, AN/VRC-43 through AN/VRC-49, AN/VRC-53, AN/VRC-64, AN/GRC-125, AN/GRC-160 and AN/GRC-163.

6.2 Ordering Data. Procurement documents should specifiy the following:

a. Title, number, and date of this specification and any amendment thereto, together with issue letters of Drawing and Data Lists cited in 3.2.

b. Level A or B preservation and packing (see section 5).

c. When rough handling test is required (see section 4).

d. Marking and shipping of samples.

e. Place of final inspection.

6.3 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action.

6.4 Location of Seal Pressure Test. It is desirable that the seal pressure test (4.13) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the seal pressure test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AS-2731W/GRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification Inspection. Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Theory of Operation. Antenna AS-2731()/GRC is electrically equivalent to an asymmetrically-fed vertical dipole of constant length whose current distribution throughout the operating band is optimized by a preselected termination within the matching unit base. The upper 3/4 of the dipole is a straight conductor with top loading. The lower 1/4 of the dipole is the outside surface of a coaxial cable which is terminated, at the bottom end, against ground in a controlled admittance. The location of the feed-point of the antenna is where the upper 3/4 of the dipole is joined to the lower 1/4 of the dipole. The coaxial cable making up the lower 1/4 of the dipole has two functions. It serves to

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supply RF energy to the feed-point and also serves as a coaxial transformer section grouping the input impedance with a 3.5:1 VSWR. The termination within the matching unit is a ferrite core RF choke whose susceptance to ground is changed (in ten sub-bands) by alternately shunting the sub-bands with preset variable components. The shunting is accomplished by a step-action rotary solenoid driven switch, which is controlled by a master wafer built into Radio Set AN/VRC-12. A manual override knob serves to indicate the sub-band in operation and provides a capability for manual selection of the desired sub-band when the antenna is used with a radio set other than those listed in 6.1.

6.8 Antenna Alignment Cavity MX-9943/GRC. This cavity is the property of the US Army Communications Electronics Command. It is designed for accurate and consistent adjustment of base admittance for the MX-9146/GRC. The alignment cavity can be obtained from the US Army Communications Electronics Command, Fort Monmouth, New Jersey.

6.9 Alignment Cavity Control C-10349/GRC. This control box is the property of the US Army Communications Electronics Command. It is designed to automatically supply operating voltages to the MX-9146/GRC and the alignment cavity in each switch operating position. The control box can be obtained from the US Army Communications Electronics Command, Fort Monmouth, New Jersey.

6.10 Antenna Test Chamber MX-9930/GRC. This chamber is the property of the US Army Communications Electronics Command. It is a cylindrical chamber which provides a capability to simplify and expedite power testing of the electrical components of the AS-2731/GRC. This chamber can be obtained from the US Army Communications Electronics Command, Fort Monmouth, New Jersey.

6.11 Definitions.

6.11.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.11.2 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.11.3 Microbial Growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.11.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.11.5 Tubular Germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-A-49263(CR) Antenna AS-2731()/GRC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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